

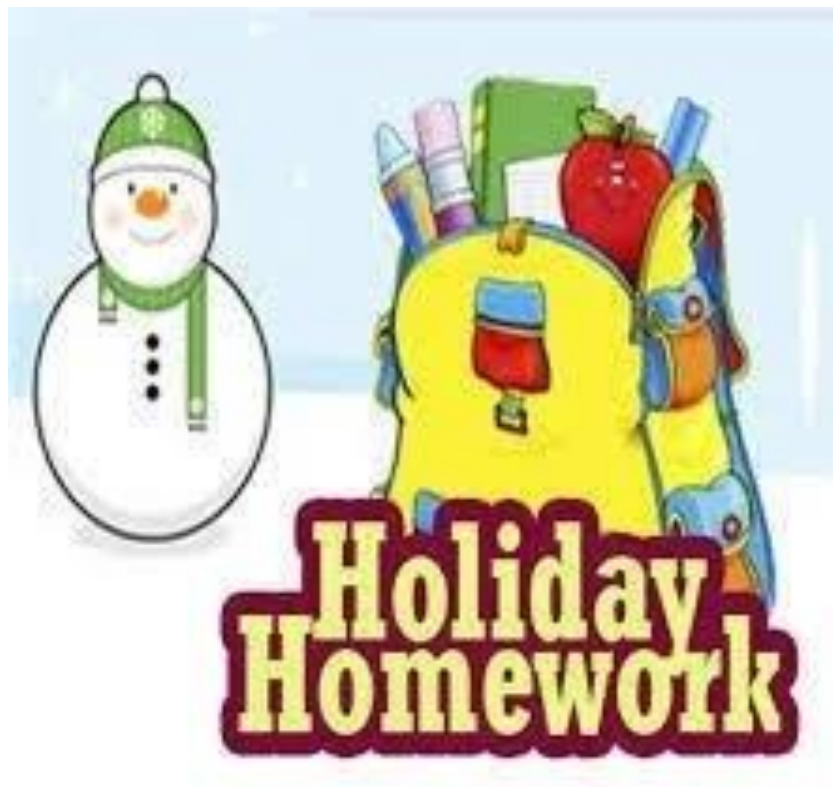


G.D. Goenka Public School, Aligarh

Winter Holiday homework

Session: 2025 – 26

Class – 9



Name of Student:

HURRAY! IT'S TIME FOR FUN AND PLAY!

IT'S TIME FOR HOLIDAYS!!

Shorter days and longer nights, Brighter lights and dark shades.

Favourite music, best friends near, Smiles and laughter everywhere.

Keep away pencils, keep away pens, Books can rest till school begins.

Make these days the time of your life, Make the nights feel just as right.

This time comes only once a year, so live it up—no stress, no fear.

This winter break, go make a change, learn, explore, grow, and rearrange.

Instructions for the parents:

- Winter Vacation start from **27 December'25 to 10th January'26**
- Spend some quality time with your kids.
- Have at least one time meal with them. Play at least one indoor or outdoor game with them.
- Take them to the park and share your childhood experiences with them.
- Encourage your child to go outdoors and spend time with their grandparents rather than sitting in front of electronic gadgets all day.
- Encourage your child to cultivate the reading habit. Get some story books of their choice to read.
- Clear all dues up to 3rd quarter (Oct-Dec) till 31 December 2025. Otherwise late fine will be charge.
- Submit admission related documents like DOB / TC / Aadhar card, etc on top priority.
- School office will be in operational mode during the vacation
- We admire the frequent use of school ERP / school whatsapp as a formal source of school information.
- Feel free to call on 8126747489 and 8265826237.

Instructions for kids:

- Do your homework neatly and on your own.
- Learn and revise all the work done in the class.
- Make your own time table for winter vacations and get it laminated. Keep proper time for playing and reading books.

For example:

Timings	Activity
8:00–8:30 am	Wake up, brush teeth, take bath
8:30 – 8:45 am	Breakfast

- Before going to bed at night, Plan your tasks for the next day and pen down them in small diary. You can write at least 2-3 lines about what you did the whole day and what you will do the next day.
- Prepare a news- paper folder, decorate it and bring all your holidays activities in it.

All the written work to be done in A4 worksheet/Note book / given sheet.

WINTER HOLIDAY ASSIGNMENTS

1. English

Eng Literature: 1. **Moments:** - Summarize the chapters from 6 to 10 in your own words.

2. **Beehive:** - Write poetic device which are used in poems from 5 to 7.

Activity: Complete a project file paste pictures of poets who are in your syllabus and write one or two pages about her / him..

2. Hindi

पुस्तक – स्पर्श

1. पाठ – शुक्र तारे के समान

प्रश्न : “शुक्र तारे के समान” पाठ में लेखक ने किन गुणों को आदर्श माना है? इन गुणों का हमारे जीवन में क्या महत्व है? पाठ के आधार पर एक अनुच्छेद लिखिए।

2. पाठ – अग्निपथ

प्रश्न : “अग्निपथ” कविता का मूल संदेश क्या है?

यह कविता युवाओं को किस प्रकार प्रेरित करती है? अपने शब्दों में एक अनुच्छेद लिखिए।

3. पाठ – खुशबू रचते हैं हाथ

प्रश्न : “खुशबू रचते हैं हाथ” पाठ में श्रमिकों और कारीगरों के जीवन का चित्रण किया गया है। उनके परिश्रम और योगदान पर आधारित एक अनुच्छेद लिखिए।

 पुस्तक – संचयन

1. पाठ – कल्लू कुम्हार की उन कोटी

प्रश्न : कल्लू कुम्हार के व्यक्तित्व और कला प्रेम का वर्णन कीजिए। यह पाठ हमें क्या सीख देता है? अपने शब्दों में एक अनुच्छेद लिखिए।

 रचनात्मक लेखन

विषय – मेरा छोटा सा निजी पुस्तकालय

प्रश्न : अपने निजी पुस्तकालय का वर्णन कीजिए। उसमें रखी पुस्तकों के प्रकार, उनका महत्व और पुस्तकों से आपको मिलने वाली प्रेरणा के बारे में एक अनुच्छेद लिखिए।

 निर्देश :

प्रत्येक उत्तर 80-100 शब्दों में लिखें।

साफ-सुथरी लिखावट और शुद्ध भाषा का ध्यान रखें।
कार्य एक अलग कॉपी में करें।

3. Maths

1. **Text Book** Practice Exercise of chapters—Quadrilaterals, Surface Area and volume, Triangles

2. **Project:**

Make a chart on triangles and congruence of triangles from your book.

OR

Model of 3D shapes also write the formula of surface area and volume

4. Social Science

1. **History**

MAKE A PROJECT ON: FOREST SOCIETY AND COLONIALISM

INDEX

Introduction

Meaning of Forest Society

Forests before Colonial Rule

British Forest Policies

Scientific Forestry

Impact on Forest Dwellers

Tribal Resistance Movements

Bastar Rebellion (Case Study)

Impact of World War I on Forests

Conclusion

Bibliography

2. **Geography**

Ch 4 Climate

Case Study / Activity Question

Study the climate of any one region of India (Rajasthan / Kerala / Assam / Ladakh) and describe:

Temperature

Rainfall

Seasonal conditions

Map Work (Compulsory)

On an outline map of India, show and label:

Areas of heavy rainfall

Areas of low rainfall

Direction of South-West Monsoon winds

3. Economics

A,Q1. Write short notes on any three food security programmes:

Mid-Day Meal Scheme

Integrated Child Development Services (ICDS)

Antyodaya Anna Yojana

Pradhan Mantri Garib Kalyan Anna Yojana

Q2. What are the major challenges to food security in India?

Explain problems such as climate change, wastage of food, and corruption.

Q3. How can food security be improved in India?

Suggest four measures to strengthen food security.

Q4. Value-Based Question

“Food security ensures not only survival but also dignity of life.”

Explain this statement in 120–150 words.

B. PROJECT-BASED ACTIVITY QUESTION

Q5 . **Case Study / Activity (Compulsory)**

Study food security in any one area (your village / city / state) and answer:

Main food crops consumed

Availability of ration shops

Problems faced by people

Suggestions for improvement

(You may include an interview, pictures, or newspaper cuttings.)

4. Political science

PROJECT ON ELECTORAL POLITICS

INDEX

Introduction

Meaning of Electoral Politics

Need for Elections in a Democracy

Types of Elections in India

Electoral Process in India

Role of the Election Commission of India

Model Code of Conduct

Importance of Voters and Voting

Electoral Reforms

Conclusion

Bibliography

5. Science

Biology

A. Draw neat and well-labelled diagrams of the different types of epithelial tissues. Also, mention the location and functions of each type.

B. Compare plant tissues and animal tissues on the basis of their structure and function.

Chemistry

Prepare a table showing the elements from atomic number 1 to 30, along with their atomic masses, electronic distribution according to Bohr's model and valency.

Physics

Questions for project file:

1. Write about the Gravitation.
2. State the universal law of Gravitation.
3. Write the importance of universal law of gravitation.
4. Write about the Centripetal force.
5. Write about the acceleration due to gravity.
6. Calculate the value of g on earth.
7. Write about thrust and pressure.
8. Define buoyancy.
9. State Archimedes' principle.
10. Differentiate between mass and weight.

Note:

1. Complete above given work neatly in the project file
2. Do the question answer of the Chapter Gravitation from the given attached pdf in physics notebook.

Very Short Answer Type Questions

1. What is the value of gravitational constant G (i) on the earth, and (ii) on the moon?
2. Which force is responsible for the moon revolving round the earth?
3. Does the acceleration produced in a freely falling body depend on the mass of the body?
4. Name the property of earth which is responsible for extremely small acceleration being produced in it as a result of attraction by other small objects.
5. What is the acceleration produced in a freely falling body of mass 10 kg? (Neglect air resistance)
6. When an object is dropped from a height, it accelerates and falls down. Name the force which accelerates the object.

7. Give the formula for the gravitational force F between two bodies of masses M and m kept at a distance d from each other.
8. What force is responsible for the earth revolving round the sun?
9. What name has been given to the force with which two objects lying apart attract each other?
10. What type of force is involved in the formation of tides in the sea?
11. Which force is responsible for holding the solar system together?
12. What is the weight of a 1 kilogram mass on the earth? ($g = 9.8 \text{ m/s}^2$)
13. On what factor/factors does the weight of a body depend?
14. As the altitude of a body increases, do the weight and mass both vary?
15. If the same body is taken to places having different gravitational field strength, then what will vary: its weight or mass?
16. If the mass of an object be 10 kg, what is its weight? ($g = 9.8 \text{ m/s}^2$)
17. The weight of a body is 50 N. What is its mass? ($g = 9.8 \text{ m/s}^2$)
18. A body has a weight of 10 kg on the surface of earth. What will be its weight when taken to the centre of the earth?
19. Write down the weight of a 50 kg mass on the earth. ($g = 9.8 \text{ m/s}^2$)
20. If the weight of a body on the earth is 6 N, what will it be on the moon?
21. State whether the following statements are true or false:
 - (a) A falling stone also attracts the earth.
 - (b) The force of gravitation between two objects depends on the nature of medium between them.
 - (c) The value of G on the moon is about one-sixth ($1/6$) of the value of G on the earth.
 - (d) The acceleration due to gravity acting on a freely falling body is directly proportional to the mass of the body.
 - (e) The weight of an object on the earth is about one-sixth of its weight on the moon.
22. Fill in the following blanks with suitable words:
 - (a) The acceleration due to gravity on the moon is about of that on the earth.
 - (b) In order that the force of gravitation between two bodies may become noticeable and cause motion, one of the bodies must have an extremely large
 - (c) The weight of an object on the earth is about of its weight on the moon.
 - (d) The weight of an object on the moon is about of its weight on the earth.
 - (e) The value of g on the earth is about of that on the moon.
 - (f) If the weight of a body is 6 N on the moon, it will be about on the earth.

Short Answer Type Questions

23. Explain what is meant by the equation: $g = G \times \frac{M}{R^2}$
where the symbols have their usual meanings.
24. (a) What do you mean by the term 'free fall'?
(b) During a free fall, will heavier objects accelerate more than lighter ones?
25. Can we apply Newton's third law to the gravitational force? Explain your answer.
26. Give reason for the following:
The force of gravitation between two cricket balls is extremely small but that between a cricket ball and the earth is extremely large.
27. Describe how the gravitational force between two objects depends on the distance between them.

28. What happens to the gravitational force between two objects when the distance between them is:
- doubled?
 - halved?
29. State two applications of universal law of gravitation.
30. Explain why, if a stone held in our hand is released, it falls towards the earth.
31. Calculate the force of gravitation between two objects of masses 50 kg and 120 kg respectively kept at a distance of 10 m from one another. (Gravitational constant, $G = 6.7 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$)
32. What is the force of gravity on a body of mass 150 kg lying on the surface of the earth? (Mass of earth = $6 \times 10^{24} \text{ kg}$; Radius of earth = $6.4 \times 10^6 \text{ m}$; $G = 6.7 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$)
33. The mass of sun is $2 \times 10^{30} \text{ kg}$ and the mass of earth is $6 \times 10^{24} \text{ kg}$. If the average distance between the sun and the earth be $1.5 \times 10^8 \text{ km}$, calculate the force of gravitation between them.
34. A piece of stone is thrown vertically upwards. It reaches the maximum height in 3 seconds. If the acceleration of the stone be 9.8 m/s^2 directed towards the ground, calculate the initial velocity of the stone with which it is thrown upwards.
35. A stone falls from a building and reaches the ground 2.5 seconds later. How high is the building? ($g = 9.8 \text{ m/s}^2$)
36. A stone is dropped from a height of 20 m.
- How long will it take to reach the ground?
 - What will be its speed when it hits the ground? ($g = 10 \text{ m/s}^2$)
37. A stone is thrown vertically upwards with a speed of 20 m/s. How high will it go before it begins to fall? ($g = 9.8 \text{ m/s}^2$)
38. When a cricket ball is thrown vertically upwards, it reaches a maximum height of 5 metres.
- What was the initial speed of the ball?
 - How much time is taken by the ball to reach the highest point? ($g = 10 \text{ m s}^{-2}$)
39. Write the differences between mass and weight of an object.
40. Can a body have mass but no weight? Give reasons for your answer.
41. A force of 20 N acts upon a body whose weight is 9.8 N. What is the mass of the body and how much is its acceleration? ($g = 9.8 \text{ m s}^{-2}$)
42. A stone resting on the ground has a gravitational force of 20 N acting on it. What is the weight of the stone? What is its mass? ($g = 10 \text{ m/s}^2$)
43. An object has mass of 20 kg on earth. What will be its (i) mass, and (ii) weight, on the moon? ($g \text{ on moon} = 1.6 \text{ m/s}^2$)
44. Which is more fundamental, the mass of a body or its weight? Why?
45. How much is the weight of an object on the moon as compared to its weight on the earth? Give reason for your answer.



It is quite difficult to lift heavy weights on the earth but it becomes very easy to lift the same heavy weights on the moon. Why?

Long Answer Type Questions

46. (a) Define mass of a body. What is the SI unit of mass?
 (b) Define weight of a body. What is the SI unit of weight?
 (c) What is the relation between mass and weight of a body?
47. (a) State the universal law of gravitation. Name the scientist who gave this law.
 (b) Define gravitational constant. What are the units of gravitational constant?
48. (a) What do you understand by the term 'acceleration due to gravity of earth'?
 (b) What is the usual value of the acceleration due to gravity of earth?
 (c) State the SI unit of acceleration due to gravity.
49. (a) Is the acceleration due to gravity of earth 'g' a constant? Discuss.
 (b) Calculate the acceleration due to gravity on the surface of a satellite having a mass of 7.4×10^{22} kg and a radius of 1.74×10^6 m ($G = 6.7 \times 10^{-11}$ Nm²/kg²). Which satellite do you think it could be?
50. The mass of a planet is 6×10^{24} kg and its diameter is 12.8×10^3 km. If the value of gravitational constant be 6.7×10^{-11} Nm²/kg², calculate the value of acceleration due to gravity on the surface of the planet. What planet could this be?

Multiple Choice Questions (MCQs)

51. An object is thrown vertically upwards with a velocity u , the greatest height h to which it will rise before falling back is given by:
 (a) u/g (b) $u^2/2g$ (c) u^2/g (d) $u/2g$
52. The mass of moon is about 0.012 times that of earth and its diameter is about 0.25 times that of earth. The value of G on the moon will be:
 (a) less than that on the earth (b) more than that on the earth
 (c) same as that on the earth (d) about one-sixth of that on the earth
53. The value of g on the surface of the moon:
 (a) is the same as on the earth (b) is less than that on the earth
 (c) is more than that on the earth (d) keeps changing day by day
54. The atmosphere consisting of a large number of gases is held to the earth by:
 (a) winds (b) clouds
 (c) earth's magnetic field (d) gravity
55. The force of attraction between two unit point masses separated by a unit distance is called:
 (a) gravitational potential (b) acceleration due to gravity
 (c) gravitational field strength (d) universal gravitational constant
56. The weight of an object at the centre of the earth of radius R is:
 (a) zero (b) R times the weight at the surface of the earth
 (c) infinite (d) $1/R^2$ times the weight at the surface of the earth
57. Two objects of different masses falling freely near the surface of moon would:
 (a) have same velocities at any instant (b) have different accelerations
 (c) experience forces of same magnitude (d) undergo a change in their inertia
58. The value of acceleration due to gravity of earth:
 (a) is the same on equator and poles (b) is the least on poles
 (c) is the least on equator (d) increases from pole to equator

59. The law of gravitation gives the gravitational force between:
- (a) the earth and a point mass only (b) the earth and the sun only
(c) any two bodies having some mass (d) any two charged bodies only
60. The value of quantity G in the formula for gravitational force:
- (a) depends on mass of the earth only (b) depends on the radius of earth only
(c) depends on both mass and radius of earth (d) depends neither on mass nor on radius of earth
61. Two particles are placed at some distance from each other. If, keeping the distance between them unchanged, the mass of each of the two particles is doubled, the value of gravitational force between them will become:
- (a) $1/4$ times (b) $1/2$ times (c) 4 times (d) 2 times
62. In the relation $F = G \times M \times m/d^2$, the quantity G :
- (a) depends on the value of g at the place of observation
(b) is used only when the earth is one of the two masses
(c) is the greatest on the surface of the earth
(d) is of the same value irrespective of the place of observation
63. The gravitational force of attraction between two objects is x . Keeping the masses of the objects unchanged, if the distance between the objects is halved, then the magnitude of gravitational force between them will become:
- (a) $x/4$ (b) $x/2$ (c) $2x$ (d) $4x$
64. An apple of mass 100 g falls from a tree because of gravitational attraction between the earth and the apple. If the magnitude of force exerted by the earth on the apple be F_1 and the magnitude of force exerted by the apple on the earth be F_2 , then:
- (a) F_1 is very much greater than F_2 (b) F_2 is very much greater than F_1
(c) F_1 is only a little greater than F_2 (d) F_1 and F_2 are exactly equal

G.D. GOENKA PUBLIC SCHOOL, ALIGARH



PRACTICAL FILE

CLASS IX

INFORMATION TECHNOLOGY

CODE : 402

SESSION : 2025-26

SUBMITTED BY:

STUDENT NAME :

ROLL NO. :

CLASS & SECTION :

SUBMITTED TO:

TEACHER'S NAME

INDEX

Sr. No.	NAME OF PRACTICAL
1.	
2.	
3.	
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INFORMATION TECHNOLOGY

SUBJECT CODE-402

CLASS-IX

PRACTICAL FILE ASSIGNMENT

Note:- Following assignments to be done in practical File

Q1. Create a poster for “Air Pollution” in MS-Word with the following features:

1. Font face
2. Font size
3. Font color
4. Header & Footer
5. Images

Q2. Create a list of 10 students by inserting a table in Ms-Word with the following fields:

1. Sr. No.
2. Student Name
3. Father’s Name
4. City

Q3. Create an article on “Gandhi & New India” using maximum feature of MS-Word.

Q4. Design a marksheet for a student as shown in figure below and create column chart based on the below data:

	A	B	C	D	E	F	G	H
1								
2	S.No.	Name	English	Hindi	Maths	Science	Total	
3	1	Abhas	89	76	93	91	349	
4	2	Anita	92	82	98	91	363	
5	3	Raja	95	81	76	82	334	
6	4	Subbu	84	72	99	94	349	
7	5	Priya	94	93	100	97	384	
8	6	Gautam	78	89	69	72	308	

Q5. Create a presentation (4-6 slides) on any one of the following topics.

1. Air Pollution
2. Traffic Rules
3. Water Conservation

Note:-Take the print of all slides.